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COMMERCIAL ITEM DESCRIPTION

# CLOTH, POPLIN: POLYESTER AND COTTON (WATER REPELLENT)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers the requirements for a polyester and cotton, water repellent, poplin cloth used on items intended for wear by military personnel of the Department of Defense.

2. SALIENT CHARACTERISTICS.

2.1 Material description. The cloth shall be a blend of 65±3 percent polyester and 35±3 percent carded and combed cotton. The blended yarns shall be spun into either a single or 2-ply yarn for the warp direction and a single yarn for the filling direction. The cloth shall be a plain weave. The cloth shall be treated with a water repellent treatment.

2.2 Color. The color of the finished cloth shall be as specified in the contract or purchase order (see 6.4) and shall meet the colorfastness requirements as specified in table I. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyed cloth shall be free of labile sulfur when tested as specified in 2.2.1.

2.2.1 Visual shade matching. The color and appearance of the water repellent-treated material shall match the standard sample when viewed using AATCC EP9, Visual Assessment of Color Difference of Textiles, option C (see 6.5), with sources simulating artificial daylight D65 illuminant (see 6.5.1) with a color temperature of 6500±200 Kelvin (K) illumination of 100±20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856±200K.

2.3 Physical requirements. The water repellent finished cloth shall conform to the physical requirements specified in table I when tested in accordance with the methods cited in table I.

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FSC 8305

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TABLE I. Physical requirements.

| Characteristic | Requirement | Test Method |
|---|---|---|
| Fiber content, percent | | |
| Polyester | 65±3 | AATCC TM20A (see 6.4) |
| Cotton | 35±3 | |
| Ply | | |
| Warp | Single or 2-ply yarn | Visual [1/] |
| Filling | Single yarn | Visual [1/] |
| Weave | Plain | Visual [1/] |
| Weight, oz/yd² | 5.3 to 6.8 | ASTM D3776/D3776M, option C |
| Yarns per inch (min.) | | |
| Warp | 108 | ASTM D3775 |
| Filling | 68 | |
| Air permeability, ft³/min/ft² (max.) | 10 | ASTM D737 |
| Breaking strength, pounds (min.) | | |
| Warp | 185 | ASTM D5034 |
| Filling | 70 | |
| Tearing strength, pounds (min.) | | |
| Warp | 5.0 | ASTM D1424 |
| Filling | 2.0 | |
| Dimensional change (after 3 cycles), percent (max.) | | |
| Warp | ±2.0 | AATCC TM135 (1)(IV)(Aiii) [2/] |
| Filling | ±2.0 | |
| Colorfastness (min.) | | |
| Laundering (after 3 cycles) | | AATCC TM61 test 2A [3/, 4/] |
| Color change and staining | 3-4 | |
| Perspiration (acid and alkaline) | | AATCC TM15 [4/] |
| Color change and staining | 4 | |
| Crocking (wet and dry) | 2.5 | AATCC TM8 [5/] |
| Dry cleaning | | AATCC TM132 [4/] |
| Color change and staining | 3-4 | |
| Lightfastness (after 40 AFU or 170 kJ/[m²nm] at 420 nm) | 4 | AATCC TM16.3 option 3 [6/] |
| pH | 5.0 to 8.5 | AATCC TM81 |
| Spray rating (min.) | | |
| Initial | 100, 100, 90 | AATCC TM22 |
| After 1 dry cleaning | 70, 70, 70 | AATCC TM22 and AATCC TM86 |



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TABLE I. Physical requirements – Continued.

| Characteristic | Requirement | Test Method |
|---|---|---|
| Dynamic absorption, percent (max.) | | |
| Initial | 18 | AATCC TM70 |
| After 3 launderings | 18 | AATCC TM135 (1)(IV)(Aii) and AATCC TM70 2/ |
| Hydrostatic pressure, cm (min.) | | |
| Initial | 30 | AATCC TM127 |
| After 3 launderings | 25 | AATCC TM135 (1)(IV)(Aiii) and AATCC TM127 2/ |
| After 1 dry cleaning | 25 | AATCC TM158 and AATCC TM127 |
| Resistance to organic liquid (min.) | | |
| Initial | No wetting to N-Tetradecane | AATCC TM118 |
| After 3 launderings | No wetting to N-Tetradecane | AATCC TM135 (1)(IV)(Aiii) and AATCC TM118 2/ |
| Labile sulfur | "Free" | 2.3.1 |
| Toxicity | 2.4 | 2.4.1 |

FOOTNOTES:

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

2/ Specimens shall be subjected to three complete cycles (wash and dry) prior to determination of dimensional change, dynamic absorption, hydrostatic height, and resistance to organic liquid after laundering.

3/ For the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated.

4/ Rated using AATCC EP1, Gray Scale for Color Change, and AATCC EP2, Gray Scale for Staining.

5/ Rated using AATCC EP8, 9-Step Chromatic Transference Scale.

6/ Rated using AATCC EP1.

2.3.1 Presence of labile sulfur. In the determination of labile sulfur in textile materials with lead acetate, two 1.50±0.01-gram samples from each material submitted for evaluation shall be tested. Each of the two samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of American Chemical Society (ACS) grade stannous chloride crystals in 100 milliliters of ACS grade hydrochloric acid (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution shall contain 5.0 grams of chemically pure (CP) reagent grade lead acetate and enough distilled water to make up a 100-milliliter solution; if the solution is not clear, a few drops (one at a time) of glacial acetic acid shall be added until the solution is clear. The test tube containing the textile sample, stannous chloride, and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

a. Free – the filter paper shows no discoloration or staining of any kind.
b. Slight – the filter paper shows a light tan to light brown discoloration stain.
c. Moderate – the filter paper shows a dark brown discoloration stain.
d. Severe – the filter paper shows a black discoloration stain.

The rating shall be recorded. The results shall be recorded as "pass" or "fail".



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2.4 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 2.4.1. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. When specified (see 6.4), the contractor shall furnish information that certifies that the finished product is composed of materials that have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact.

2.4.1 Toxicity assessment. When specified (see 6.4), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of the studies indicate the cloth is not a sensitizer or irritant, a human Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure in "Hayes' Principles and Methods of Toxicology." If the toxicity requirement (see 2.4) can be demonstrated with historical data on the finished treatments used, then toxicity testing may not be required.

2.5 Labels. Each roll of finished cloth shall be labeled or ticked for fiber content in accordance with 16 CFR § 303, Rules and Regulations Under the Textile Fiber Products Identification Act.

2.6 Workmanship. The finished cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the technical data in the bid package.

3. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The finished cloth provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The finished cloth must have been sold to the commercial market or the government.

4.3 Visual examination. The finished cloth shall be examined for the following defects as defined in ASTM D3990, Standard Terminology Relating to Fabric Defects:

a. Cloth – any spot or stain more than 1 inch in its longest dimension; any hole, cut, or tear; a missing end or pick; thick or thin areas; overall uncleanliness; a crease or hard crease; slubs, knots, slugs, or gout; and shade bar, streaks, or uneven dying. Water repellent treatment not uniform.

b. Label – label missing, incorrect, or illegible. Required information missing from the label.

c. Packaging – not packaged in accordance with the contract or purchase order (see 6.4).

4.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 6.4).

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Intended use. This cloth is intended to be used in items with a water-repellency requirement.

6.2 Source of documents.

6.2.1 American Association of Textile Chemists and Colorists (AATCC). AATCC test methods are available from www.aatcc.org or P.O. Box 12215, Research Triangle Park, NC 27709.

6.2.2 ASTM International. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428.

6.2.3 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov or the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401.



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6.2.4 Hayes' Principles and Methods of Toxicology. Hayes' Principles and Methods of Toxicology is available from www.crcpress.com or may be available from other retailers.

6.3 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirements.

6.4 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Color required (see 2.2).
c. When toxicity testing is required (see 2.4 and 2.4.1).
d. Product conformance provisions (see 4.1).
e. Packaging requirements (see 4.3.c and 5).
f. Acceptance criteria provisions (see 4.4).

6.5 Visual shade matching. In 2019, option A of AATCC EP9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35±5-degree angle relative to the horizontal. The observer will view the specimens at a 90-degree angle, relative to the plane of the specimens.

6.5.1 Use of D75 illuminant. The use of D75 illuminant with a color temperature of 7500±200 Kelvin (K) illumination of 100±20 foot candles in lieu of the specified D65 illuminant is permitted.

6.6 Key words.

Ply
Raincoat
Single

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – GL
Navy – MC

Review Activity:
DLA – CT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – MC
(Project 8305-2021-027)



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